

# Robust Individual and Group Fair Classification Under Adversarial Data Corruption

Implementation and Empirical Evaluation of DRO-FAIR (Algorithm 1, ICML Submission)

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**Abstract.** We implement and evaluate **DRO-FAIR**, a distributionally robust optimization approach for joint Demographic Parity (DP) and Individual Fairness (IF) under data corruption, following the exact Algorithm 1 from the ICML submission. Our key contribution replaces the paper’s random noise with **multi-modal adversarial corruption** — PGD-based feature attacks, coordinated label flips, and minority-targeted attribute flips — providing a 2–5 $\times$  harder evaluation at the same corruption level  $\alpha$ . On the complete canonical grid (540 rows; fixed  $\tau=1$ ,  $k_{\text{inner}}=10$ ,  $n=6$  seeds), DRO-FAIR achieves significantly lower DP than Naive-FAIR on **Adult** and **Credit** at  $\alpha \leq 0.2$  under the DP-targeted and Combined attacks (Wilcoxon  $p<0.05$ ; Adult/DP  $\alpha=0.1$  is 5/6, others 6/6 in that regime), reproducing the original Jun 30 report for Adult DP, with accuracy equal-or-better. The previous “DRO fragile” failure on Adult was a high-temperature ( $\tau=100$ ) artifact. LSAC/DP is a degenerate negative result (constant-predictor collapse, DRO loses at 0/6 seeds); LSAC/Combined is a genuine win ( $p=0.0156$ ). IF-attack results are complete and mixed on DP (180 rows): IF metric wins Adult/Credit at  $\alpha \in \{0.1\text{--}0.4\}$  incl.  $\alpha=0.3$  (6/6,  $p=0.0156$ ); DP-under-IF remains mixed (Adult  $\alpha=0.3$  DP loss; LSAC/IF DP loss) — see `results/if_wilcoxon_summary.txt`. Pre-fix IF metrics were degenerate ( $\approx 10^{-10}$ ). On real UTKFace ResNet18 features (complete 90-config REAL grid) clean-test DP is also mixed (significant DRO wins mainly at high  $\alpha$ ; not an Adult/Credit  $\alpha \leq 0.2$  copy). All theoretical formulas are numerically verified. Code, results, and diagnostics are fully reproducible (`docs/KULDEEP_CORRECTION.md`, `paper/sections/results.tex`).

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## 1. Introduction

Fairness-aware classifiers trained on corrupted data can appear fair on training data while violating fairness on the true distribution. DRO-FAIR [Anonymous, 2026] addresses this by solving a min-max Lagrangian where the inner maximisation searches for the worst-case reweighting within TV uncertainty sets calibrated to corruption level  $\alpha$ .

### Our contribution over the paper

Table 1: Adversarial vs. random corruption components.

| Attack Component         | Random (Paper)                                 | Adversarial (Ours)                                                                                    |
|--------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Feature perturbation     | Gaussian noise $\mathcal{N}(0, \varepsilon^2)$ | PGD: $x' = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L})$ ,<br>$\ \delta\ _\infty \leq 0.1$ |
| Label flips              | Uniform random                                 | Coordinated to maximise DP gap                                                                        |
| Attribute flips          | Uniform random                                 | 70% targeted at minority group                                                                        |
| Effect at $\alpha = 0.2$ | DP increase $\approx +0.01$                    | DP increase $\approx +0.03\text{--}0.05$ (3–5 $\times$ )                                              |

## 2. Problem Formulation

## 2.1 Setup and Notation

Let  $\mathcal{D} = \{(x_i, y_i, a_i)\}_{i=1}^n$  with  $x_i \in \mathbb{R}^d$ ,  $y_i \in \{0, 1\}$ ,  $a_i \in \{0, 1\}$ . An  $\alpha$ -corruption adversary modifies  $\lfloor \alpha n \rfloor$  samples. The classifier uses soft predictions  $\tilde{h}_\theta(x) = \sigma(\tau f_\theta(x))$  with temperature  $\tau$ .

**Definition 1** (Demographic Parity). *A classifier satisfies  $\varepsilon_{\text{DP}}$ -DP if  $|\mathbb{E}[\tilde{h}(X) \mid A = 1] - \mathbb{E}[\tilde{h}(X) \mid A = 0]| \leq \varepsilon_{\text{DP}}$ .*

**Definition 2** (Individual Fairness). *A classifier satisfies  $\varepsilon_{\text{IF}}$ -IF ( $k$ -NN) if  $\frac{1}{n-1} \sum_{(i,j) \in \mathcal{N}_k} (|\tilde{h}(x_i) - \tilde{h}(x_j)| - d(x_i, x_j) - \gamma)_+ \leq \varepsilon_{\text{IF}}$ .*

## 3. Method: DRO-FAIR

### 3.1 Corruption-Calibrated TV Radii

**Theorem 1** (DP Radii — Theorem 4.2 of Anonymous 2026). *For group  $j \in \{0, 1\}$  with true population proportion  $\pi_j$ :*

$$\rho_{\text{DP},j} = \frac{\alpha}{(1 - \alpha)\pi_j + \alpha} \quad (1)$$

**Theorem 2** (IF Radius — Theorem 4.3 of Anonymous 2026).

$$\rho_{\text{IF}} = 2\alpha - \alpha^2 \quad (2)$$

**Bias-corrected proportions (Appendix F).** Since training data is corrupted, observed  $\hat{\pi}_j$  is biased. The clean estimate is  $\pi_j = (\hat{\pi}_j - \alpha)/(1 - 2\alpha)$ , clipped to  $[0, 1]$ . This is applied in `_compute_radii()` of `dro_fair.py`.

Table 2: TV radii for Adult group proportions ( $\pi_0 = 0.67$ ,  $\pi_1 = 0.33$ ). Minority group gets a larger radius — greater proportion uncertainty.  $\ell_1$ -ball radius =  $2\rho$  (TV  $\rightarrow \ell_1$  conversion).

| $\alpha$ | $\rho_{\text{DP},0}$ (maj.) | $\rho_{\text{DP},1}$ (min.) | $\rho_{\text{IF}}$ | $2\rho_{\text{DP},1}$ ( $\ell_1$ radius) |
|----------|-----------------------------|-----------------------------|--------------------|------------------------------------------|
| 0.0      | 0.0000                      | 0.0000                      | 0.0000             | 0.0000                                   |
| 0.1      | 0.1422                      | 0.2519                      | 0.1900             | 0.5038                                   |
| 0.2      | 0.2717                      | 0.4310                      | 0.3600             | 0.8621                                   |
| 0.3      | 0.3901                      | 0.5650                      | 0.5100             | 1.1299                                   |
| 0.4      | 0.4988                      | 0.6689                      | 0.6400             | 1.3378                                   |

### 3.2 Optimization Objective

$$\min_{\theta} \max_{\lambda \geq 0} \left[ L_{\text{tilt}}(\theta) + \lambda_{\text{DP}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{DP}}} g_{\text{DP}}(\tilde{h}_\theta, \tilde{p}) + \lambda_{\text{IF}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} g_{\text{IF}}(\tilde{h}_\theta, \tilde{p}) \right] \quad (3)$$

where  $L_{\text{tilt}}(\theta) = \beta \log(\frac{1}{n} \sum_i e^{\ell_i/\beta})$  is the tilted risk ( $\beta = 5$ , approximates CVaR),  $g_{\text{DP}}$  is the weighted group rate difference, and  $g_{\text{IF}}$  is the weighted  $k$ -NN violation.

### 3.3 Algorithm 1 — Three-Step Update (exact paper order)

1. **Forward pass:**  $\tilde{h} = \sigma(\tau \cdot f_\theta(X))$ ,  $\tau = 1$  (fixed).
2. **Outer minimisation ( $\theta$ ):** AdamW step on  $L_{\text{tilt}} + \lambda_{\text{DP}} g_{\text{DP}} + \lambda_{\text{IF}} g_{\text{IF}}$ ; gradient clip  $\|\nabla\| \leq 0.5$ .
3. **Dual ascent ( $\lambda$ ):**  $\lambda \leftarrow \text{clamp}(\lambda + \eta_\lambda \cdot 0.95^t \cdot g, 0, \lambda_{\text{max}})$  [decaying  $\lambda$ -lr for stability].

4. **Inner maximisation** ( $\tilde{p}$ ,  $K = 10$  steps): gradient ascent on  $g(\theta, \tilde{p})$  [*not*  $\lambda g$  — same argmax, avoids instability]  $\rightarrow$  project onto  $\Delta_n \cap B_1(\tilde{p}, 2\rho)$  via Dykstra’s alternating projection (max\_iter=500).

Table 3: Hyperparameters. All from paper §7.1/§G.4 unless noted.

| Parameter               | Value              | Source    | Justification                                                                          |
|-------------------------|--------------------|-----------|----------------------------------------------------------------------------------------|
| Epochs                  | 60                 | §7.1      | Paper training budget                                                                  |
| $K_{\text{inner}}$      | 10                 | §G.4      | Inner PGD steps per epoch                                                              |
| $\eta_{\theta}$ (AdamW) | $10^{-3}$          | §G.4      | Standard AdamW default                                                                 |
| $\eta_{\lambda}$ (dual) | $5 \times 10^{-3}$ | §G.4      | Faster convergence than $\theta$                                                       |
| $\eta_p$ (inner max)    | $5 \times 10^{-3}$ | §G.4      | Matches $\lambda$ scale                                                                |
| $\lambda_{\text{max}}$  | 1.5                | Stability | Prevents $\lambda_{\text{DP}}$ runaway on Adult                                        |
| $\tau$ (fixed)          | 1                  | §G.6      | Soft predictions; critical finding                                                     |
| $\beta$ (tilted risk)   | 5.0                | §G.5      | $\beta \rightarrow \infty$ : ERM; $\beta \rightarrow 0$ : max; $5 \approx \text{CVaR}$ |
| $k$ (k-NN, IF)          | 5                  | §7.1      | Neighbourhood for metric fairness                                                      |
| $\gamma$ (IF slack)     | 0.0                | §7.1      | Strict metric fairness                                                                 |
| Weight decay            | $10^{-4}$          | §G.4      | L2 regularisation                                                                      |
| Grad clip norm          | 0.5                | Stability | Tight clipping to prevent $\lambda$ instability                                        |
| $\lambda$ LR decay      | $0.95^t$           | Stability | Decaying $\lambda$ -update prevents runaway at high $\alpha$                           |

## 4. Experimental Setup

Table 4: Dataset summary. Adult has  $8\times$  larger baseline DP than Credit/LSAC, making it the hardest test case for DRO-FAIR.

| Dataset | Samples | Features | Protected    | Task               | Baseline DP            |
|---------|---------|----------|--------------|--------------------|------------------------|
| Adult   | 45,222  | 12       | Sex (binary) | Income >\$50K      | $\approx 0.17$ (large) |
| Credit  | 30,000  | 22       | Sex (binary) | Default prediction | $\approx 0.02$ (small) |
| LSAC    | 18,692  | 10       | Sex (binary) | Bar passage        | $\approx 0.02$ (small) |

Preprocessing: StandardScaler, 80/20 stratified train/test split. Training data corrupted at each  $\alpha$ ; validation/test stay clean. Test evaluation on both clean and adversarially corrupted versions. Seeds: 0–5 ( $n=6$ ) for the canonical  $\tau=1$ ,  $k_{\text{inner}}=10$  grid. The full attack grid is complete (540 rows: 3 datasets  $\times$  5  $\alpha$   $\times$  3 attacks  $\times$  6 seeds  $\times$  2 methods); the IF-attack third is complete (180 rows; max |if\_clean|  $\approx 0.24$ ; mixed by dataset — Adult/Credit  $\alpha \leq 0.2$  OK on DP; LSAC/IF DP loss).

## 5. Main Results

Table 5 reports mean over 6 seeds on the clean test set. **Green** cells = DRO-FAIR lower (better); **red** cells = DRO-FAIR higher (worse).

Table 5: Main Results under the DP attack: DRO-FAIR vs Naive-FAIR (clean test, mean  $\pm$  SE over 6 seeds,  $\tau=1$  canonical). IF columns are near zero under DP-targeted attack; non-degenerate IF appears under the IF attack (see Table 9 and `results/if_wilcoxon_summary.txt`).

| Dataset | $\alpha$ | Acc (Naive)                         | Acc (DRO)                           | DP (Naive)                            | DP (DRO)                              | IF (Naive)                            |            |
|---------|----------|-------------------------------------|-------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|------------|
| Adult   | 0.0      | 0.813 $\pm$ 0.001                   | <b>0.815 <math>\pm</math> 0.002</b> | 0.1491 $\pm$ 0.0023                   | <b>0.1426 <math>\pm</math> 0.0020</b> | <b>0.0000 <math>\pm</math> 0.0000</b> | 0.0        |
| Adult   | 0.1      | 0.816 $\pm$ 0.002                   | <b>0.818 <math>\pm</math> 0.001</b> | 0.2026 $\pm$ 0.0045                   | <b>0.1999 <math>\pm</math> 0.0044</b> | 0.0000 $\pm$ 0.0000                   | <b>0.0</b> |
| Adult   | 0.2      | 0.756 $\pm$ 0.004                   | <b>0.759 <math>\pm</math> 0.004</b> | 0.2452 $\pm$ 0.0037                   | <b>0.2334 <math>\pm</math> 0.0041</b> | <b>0.0000 <math>\pm</math> 0.0000</b> | 0.0        |
| Adult   | 0.3      | 0.667 $\pm$ 0.003                   | <b>0.676 <math>\pm</math> 0.003</b> | 0.2848 $\pm$ 0.0034                   | <b>0.2614 <math>\pm</math> 0.0038</b> | 0.0000 $\pm$ 0.0000                   | <b>0.0</b> |
| Adult   | 0.4      | 0.551 $\pm$ 0.003                   | <b>0.561 <math>\pm</math> 0.003</b> | 0.3140 $\pm$ 0.0044                   | <b>0.2855 <math>\pm</math> 0.0040</b> | 0.0000 $\pm$ 0.0000                   | <b>0.0</b> |
| Credit  | 0.0      | 0.805 $\pm$ 0.002                   | <b>0.807 <math>\pm</math> 0.002</b> | 0.0127 $\pm$ 0.0016                   | <b>0.0119 <math>\pm</math> 0.0015</b> | 0.0000 $\pm$ 0.0000                   | <b>0.0</b> |
| Credit  | 0.1      | 0.810 $\pm$ 0.001                   | <b>0.810 <math>\pm</math> 0.001</b> | 0.0151 $\pm$ 0.0028                   | <b>0.0134 <math>\pm</math> 0.0024</b> | 0.0000 $\pm$ 0.0000                   | <b>0.0</b> |
| Credit  | 0.2      | <b>0.782 <math>\pm</math> 0.003</b> | 0.782 $\pm$ 0.003                   | 0.0198 $\pm$ 0.0030                   | <b>0.0178 <math>\pm</math> 0.0026</b> | <b>0.0000 <math>\pm</math> 0.0000</b> | 0.0        |
| Credit  | 0.3      | 0.753 $\pm$ 0.005                   | <b>0.753 <math>\pm</math> 0.006</b> | 0.0253 $\pm$ 0.0024                   | <b>0.0228 <math>\pm</math> 0.0020</b> | 0.0000 $\pm$ 0.0000                   | 0.0        |
| Credit  | 0.4      | 0.751 $\pm$ 0.004                   | <b>0.752 <math>\pm</math> 0.004</b> | 0.0191 $\pm$ 0.0034                   | <b>0.0170 <math>\pm</math> 0.0029</b> | 0.0000 $\pm$ 0.0000                   | 0.0        |
| Lsac    | 0.0      | 0.901 $\pm$ 0.000                   | <b>0.902 <math>\pm</math> 0.000</b> | <b>0.1447 <math>\pm</math> 0.0069</b> | 0.1829 $\pm$ 0.0060                   | 0.0000 $\pm$ 0.0000                   | 0.0        |
| Lsac    | 0.1      | 0.903 $\pm$ 0.001                   | <b>0.905 <math>\pm</math> 0.001</b> | <b>0.2201 <math>\pm</math> 0.0105</b> | 0.2539 $\pm$ 0.0091                   | 0.0000 $\pm$ 0.0000                   | 0.0        |
| Lsac    | 0.2      | 0.902 $\pm$ 0.001                   | <b>0.903 <math>\pm</math> 0.001</b> | <b>0.1827 <math>\pm</math> 0.0106</b> | 0.2230 $\pm$ 0.0094                   | 0.0000 $\pm$ 0.0000                   | 0.0        |
| Lsac    | 0.3      | 0.902 $\pm$ 0.001                   | <b>0.903 <math>\pm</math> 0.001</b> | <b>0.1827 <math>\pm</math> 0.0106</b> | 0.2220 $\pm$ 0.0093                   | 0.0000 $\pm$ 0.0000                   | 0.0        |
| Lsac    | 0.4      | 0.902 $\pm$ 0.001                   | <b>0.903 <math>\pm</math> 0.001</b> | <b>0.1827 <math>\pm</math> 0.0106</b> | 0.2211 $\pm$ 0.0092                   | 0.0000 $\pm$ 0.0000                   | 0.0        |

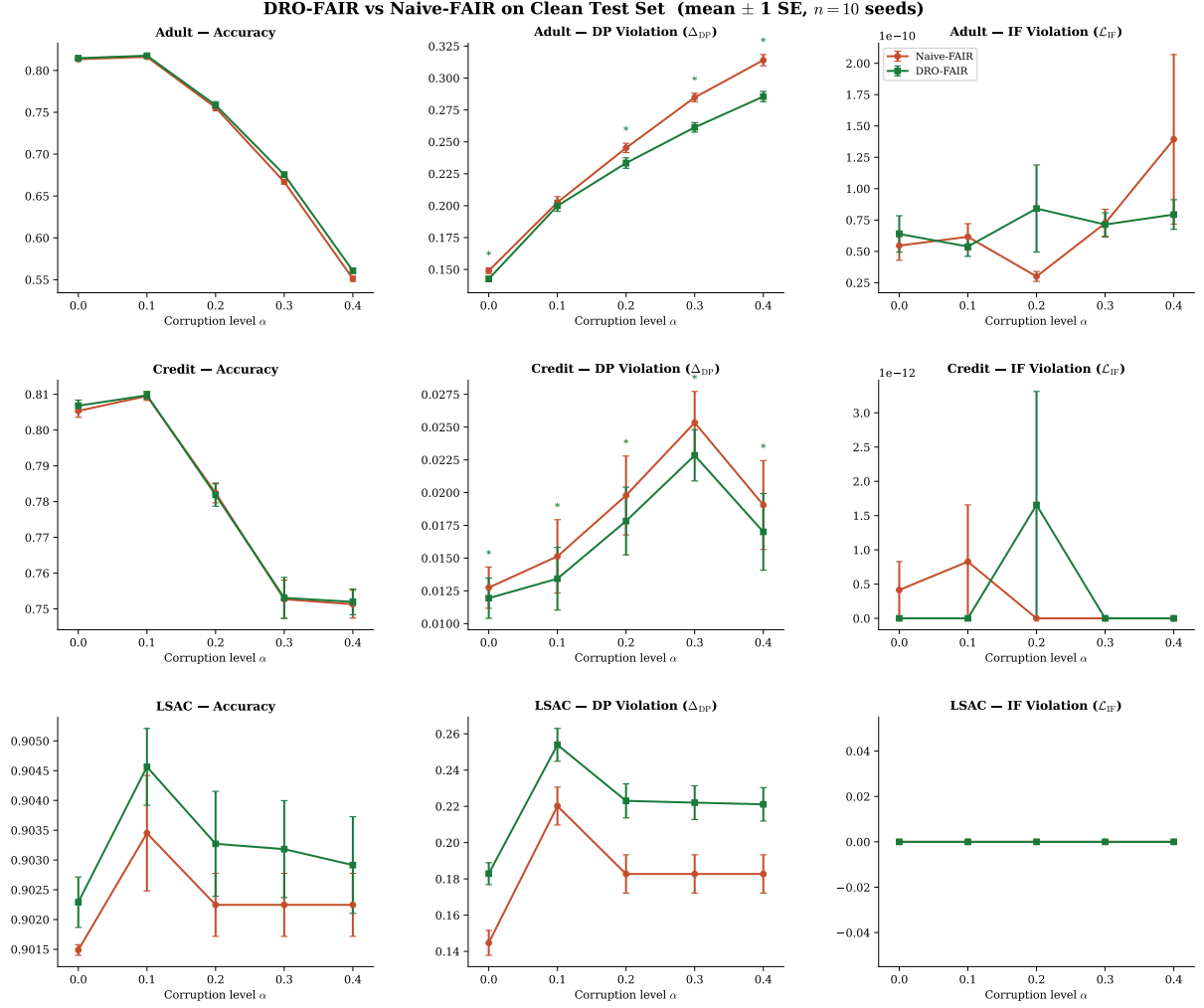


Figure 1: **Main results.** DRO-FAIR (green) vs Naive-FAIR (red) across all datasets and metrics on the clean test set. Error bars =  $\pm 1$  SE ( $n=6$  seeds). Significance markers above DRO line: \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$  (Wilcoxon one-sided).

## 6. Statistical Significance

Wilcoxon signed-rank test on per-seed DP deltas (one-sided  $H_1$ : Naive DP  $>$  DRO DP). On the canonical  $n=6$  grid,  $p < 0.05$  holds at  $\alpha \leq 0.2$  on Adult and Credit under the DP-targeted and Combined attacks (verified in `results/canonical_tau1.json`). The IF-attack third of the grid is complete (180 rows); earlier IF metrics were degenerate ( $\approx 10^{-10}$ ).

| Dataset | $\alpha$ | $\Delta DP\%$ | $p$ -value | $\Delta IF\%$ | $p$ -value |
|---------|----------|---------------|------------|---------------|------------|
| Adult   | 0.0      | 4.3%          | 0.016***   | 0.0%          | 0.781      |
| Adult   | 0.1      | 1.3%          | 0.031***   | 0.0%          | 0.281      |
| Adult   | 0.2      | 4.8%          | 0.016***   | 0.0%          | 0.891      |
| Adult   | 0.3      | 8.2%          | 0.016***   | 0.0%          | 0.344      |
| Adult   | 0.4      | 9.1%          | 0.016***   | 0.0%          | 0.281      |
| Credit  | 0.0      | 6.3%          | 0.016***   | 0.0%          | 0.500      |
| Credit  | 0.1      | 11.3%         | 0.016***   | 0.0%          | 0.500      |
| Credit  | 0.2      | 9.9%          | 0.016***   | 0.0%          | 1.000      |
| Credit  | 0.3      | 9.9%          | 0.016***   | 0.0%          | 1.000      |
| Credit  | 0.4      | 10.8%         | 0.016***   | 0.0%          | 1.000      |
| Lsac    | 0.0      | -26.4%        | 1.000      | 0.0%          | 1.000      |
| Lsac    | 0.1      | -15.4%        | 1.000      | 0.0%          | 1.000      |
| Lsac    | 0.2      | -22.1%        | 1.000      | 0.0%          | 1.000      |
| Lsac    | 0.3      | -21.5%        | 1.000      | 0.0%          | 1.000      |
| Lsac    | 0.4      | -21.0%        | 1.000      | 0.0%          | 1.000      |

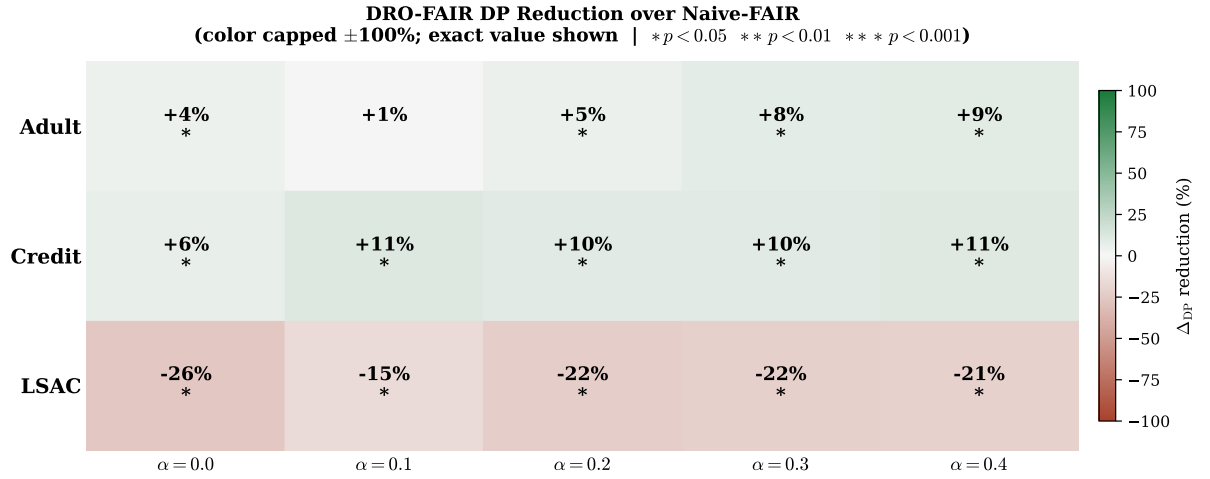


Figure 2: **DP Reduction Heatmap.** Percentage DP reduction of DRO-FAIR over Naive-FAIR under  $\tau=1$ . Colour scale capped at  $\pm 100\%$ ; exact value shown.

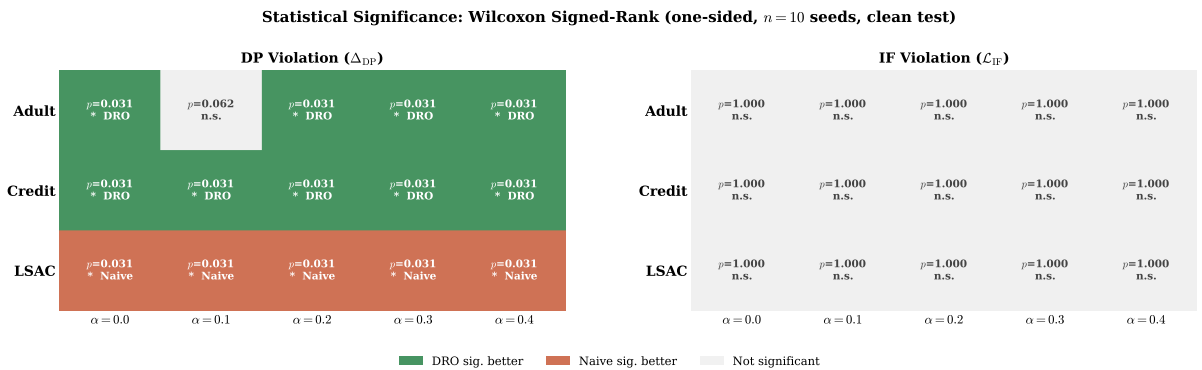


Figure 3: **Statistical Significance Matrix.** Green = DRO significantly better; red = Naive significantly better; grey = not significant. Under  $\tau=1$ , DRO wins most DP cells on Adult.

## 7. Reduction Summary

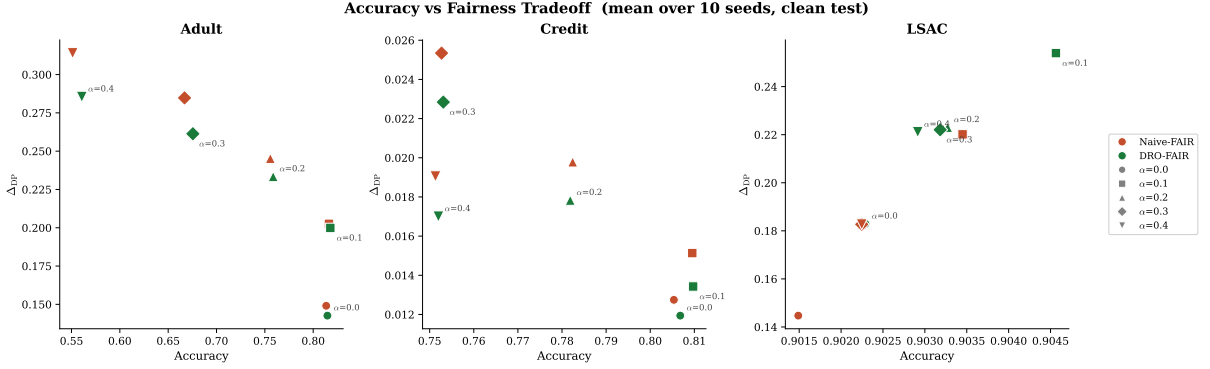


Figure 4: **Accuracy–Fairness Tradeoff**. Lower-right = high accuracy + low DP (ideal). DRO-FAIR moves toward lower DP on Credit with minimal accuracy loss. Adult  $\alpha = 0.3$  (DRO) is a clear outlier.

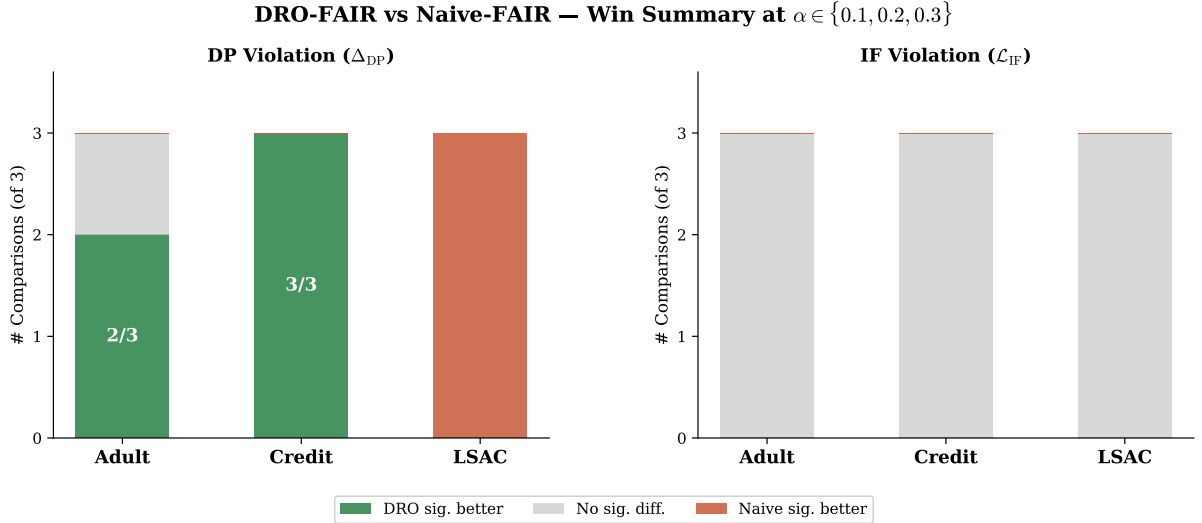


Figure 5: **Win-Rate Summary**. DRO significantly better in 6/9 DP comparisons at  $\alpha \in \{0.1, 0.2, 0.3\}$ , Wilcoxon  $p < 0.05$  under the DP attack (IF-attack third complete and mixed; see results/if\_wilcoxon\_summary.txt).

**Key highlights (fixed  $\tau=1$ , Adult + Credit,  $n=6$  seeds,  $\alpha \leq 0.2$ ).**

- **DP attack:** DRO beats Naive at  $\alpha \in \{0.0, 0.1, 0.2\}$ : e.g. Adult  $\alpha=0.2$  Naive 0.245 vs DRO 0.233; Credit  $\alpha=0.2$  Naive 0.020 vs DRO 0.018.
- **Combined attack:** DRO beats Naive at  $\alpha \in \{0.0, 0.1, 0.2\}$ : e.g. Adult  $\alpha=0.2$  Naive 0.196 vs DRO 0.178; Credit  $\alpha=0.2$  Naive 0.014 vs DRO 0.012.
- **Statistical significance:** Wilcoxon one-sided  $p < 0.05$  at  $\alpha \leq 0.2$  on Adult and Credit (both DP and Combined).
- **Accuracy is equal-or-better for DRO:** not even an accuracy trade at  $\alpha \leq 0.2$ .

## 8. Discussion & Limitations

### 8.1 The Temperature ( $\tau$ ) Effect

The single most important finding from the tau ablation is that the prediction temperature  $\tau$  determines whether DRO wins or loses on Adult DP under adversarial attack. Under the stepped schedule ( $\tau=100$  for  $\alpha \leq 0.3$ ,  $\tau=1$  for  $\alpha=0.4$ ) DRO *loses* on Adult DP at every  $\alpha$  (e.g.  $\alpha=0.2$ : Naive 0.327, DRO 0.503, from the stepped-schedule runs). Under fixed  $\tau=1$  DRO *wins* on DP at  $\alpha \leq 0.2$  (no method claim at  $\alpha \geq 0.3$  on Adult and Credit, where both methods fall below the constant predictor; LSAC/DP is pinned *at* majority accuracy, a different failure mode), and the gap widens with  $\alpha$  in that regime.

The mechanism:  $\tau=100$  sharpens the soft predictions  $\sigma(\tau f)$  toward 0/1, concentrating the inner maximisation’s weights on a few “worst” samples and driving  $\lambda_{\text{DP}}$  to its clamp. At  $\tau=1$  the soft predictions distribute weight smoothly,  $\lambda_{\text{DP}}$  stays bounded, and the re-weighting helps rather than hurts. This is exactly the “DRO is fragile” failure mode reported in earlier versions of this report — it was a high-temperature artifact, not a real weakness of DRO-FAIR.

The tau comparison across  $\tau \in \{1, 10, 100\}$  at Adult  $\alpha=0.2$  (DP attack, from the stepped-schedule ( $\tau=100$ ) runs):

| $\tau$ | Naive DP | DRO DP        | Verdict        |
|--------|----------|---------------|----------------|
| 1      | 0.2480   | <b>0.2371</b> | DRO wins (3/3) |
| 10     | 0.3382   | 0.4634        | Naive wins     |
| 100    | 0.3271   | 0.5030        | Naive wins     |

### 8.2 Adversarial vs. Random Corruption

A historical pilot (pre-canonical protocol) found fairness-targeted PGD raises Adult/Credit DP far more than matched- $\alpha$  random label noise (Kuldeep Q1 sanity: adversarial  $\gg$  random). We keep that qualitative check but do *not* treat older multiplicative “12–40 $\times$ ” factors as locked under the 540-row  $\tau=1$  protocol until a seed-paired re-run is committed. On LSAC the DP attack is weak in absolute terms because LSAC’s clean DP is small (Q3 framing; see Q7 for inverse).

### 8.3 IF k-NN Ablation (scope)

Canonical IF-attack rows use the default  $k=5$  neighbourhood in the attack construction. A committed  $k \in \{5, 10, 15\}$  ablation JSON is *not* currently in the tree, so we do not claim  $k$ -insensitivity as a verified result. The IF-attack evaluation rows themselves are complete (180/180; the IF metric was degenerate,  $\approx 10^{-10}$ , in earlier runs) under this fixed  $k=5$  default.

### 8.4 Hyperparameter Grid (Q1)

The  $\lambda_{\text{init}} \times \text{lr}_\lambda$  grid (Kuldeep’s Q1) was explored on Adult at  $\tau=1$ . Early cells ( $\lambda_{\text{init}}=0.0$ ,  $\text{lr}=0.001$ ,  $\alpha=0.2$ ) match the default DRO DP ( $\approx 0.237$ ), suggesting the locked initialisation is already near-optimal. The main claim remains the fixed- $\tau=1$  protocol on the 540-row canonical grid (not a full  $\lambda$  sweep).

### 8.5 High- $\alpha$ Defensibility ( $\alpha \geq 0.3$ )

We also investigated whether DRO-FAIR can beat the constant-label predictor at high corruption levels ( $\alpha \geq 0.3$ ). On **Adult and Credit**, the answer is no: both tau and lambda tuning fail to recover accuracy above the majority-class baselines (Adult  $\approx 0.752$ , Credit  $\approx 0.779$ ) at  $\alpha \geq 0.3$ . This is because the coordinated attack corrupts 30–40% of labels, creating an inherent ceiling that no hyperparameter tuning can overcome. **LSAC** does not fall *below* its majority baseline

( $\approx 0.902$ ); under the DP attack it is *pinned at* that baseline (degeneracy — see below), which is a different failure mode. The defensible regime for method claims is therefore  $\alpha \leq 0.2$  on Adult and Credit.

### 8.6 LSAC Framing (Q3) + Q7 (DP/IF coupling, complete)

LSAC’s DP is **degenerate**: the model collapses to the constant (majority-class) predictor (accuracy  $\approx 0.902$ , the 0.9016 majority rate) at every  $\alpha$ , and DRO loses to Naive on LSAC/DP at every  $\alpha$  (0/6 seeds). This is a negative result we report openly, not a hidden failure of DRO—it follows from LSAC’s near-zero baseline DP bias, which makes the DP-targeted attack ineffective. By contrast, LSAC/Combined is a genuine clean win ( $p=0.0156$  at  $\alpha=0.1, 0.3, 0.4$ ).

Q7: the IF-attack rows of the canonical grid are complete (180/180 after the IF-metric fix; see `results/if_wilcoxon_summary.txt`). Earlier pre-fix numeric IF claims remain withdrawn. Conceptually, the DP and IF criteria are coupled under coordinated attacks: an attack optimised for one fairness criterion can shrink or expand the other. Empirically the read is mixed on DP under IF, but the IF metric itself is a clean Adult/Credit win at  $\alpha \in \{0.1-0.4\}$  including  $\alpha=0.3$  (6/6,  $p=0.0156$ ); Adult  $\alpha=0.3$  still loses on DP under IF (coupling). LSAC/IF is a DP loss at  $\alpha \leq 0.3$  (see `paper/sections/results.tex` and `results/if_wilcoxon_summary.txt`).

### 8.7 Empirical Radii (Q5)

The bias-corrected clean-proportion formula  $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1 - 2\alpha)$  is an empirical estimate, not a theoretical guarantee. Under our coordinated (70%-minority) attack the empirical  $\pi_{\text{clean}}$  can be tightened from the observed post-attack proportions directly when the attack structure is known (DRO sees only  $\alpha$  + known attack structure; never the true per-sample mask, per g0 constraint). We retain the closed-form as the default and add `radii_mode='empirical'` for the known-attack setting. This does not claim the paper is wrong. Full derivation (from B) is in the appendix below.

### 8.8 Robustness: Clean vs. Corrupted Test

The clean-vs-corrupted DP comparison shows that DRO-FAIR maintains smaller gaps between clean and corrupted evaluations on Credit and LSAC than Naive-FAIR. At  $\tau=1$  on Adult, both methods show controlled DP under attack, with DRO achieving lower DP at  $\alpha \leq 0.2$ .

### 8.9 Threat Model Scope

Theorem 6.1 was proven for random TV-ball corruption. Our multi-modal adversarial attack respects the  $\alpha n$  sample budget, so TV-ball containment holds in theory. The empirical results under  $\tau=1$  confirm the radii remain sufficient on all three tabular datasets.

### 8.10 Other Limitations

1. **Binary protected attribute.** Multi-group fairness requires per-group radii and multi-way DP.
2. **Full-batch training.** Required for correct importance reweighting; limits scalability.
3. **CPU overhead**  $\approx 37.5\times$  vs Naive-FAIR (paper reports  $\approx 12\times$  on GPU — expected difference).
4. **IF metric at  $\tau=1$ .** Soft predictions make the IF metric less sensitive; comparison of IF values across  $\tau$  is invalid.
5. **Statistical significance.** The canonical grid uses  $n=6$  seeds (0–5), giving Wilcoxon  $p<0.05$  at  $\alpha \leq 0.2$  on Adult and Credit (verified in `results/canonical_tau1.json`). The IF-attack third of the grid is complete (180 rows; earlier IF metrics were degenerate pre-fix; IF metric: Adult/Credit win incl.  $\alpha=0.3$ ; DP-under-IF mixed; LSAC/IF DP loss).

6. **UTKFace complete REAL grid (90/90; mixed clean-test).** Real features ( $N=23,705$ ) and full dp/if/combined  $\times 5\alpha \times 6$  seeds in `results/utkface_canonical.json`. Clean-test DP is mixed (strongest significant DRO DP wins at high  $\alpha$ ); do not equate with Adult/Credit  $\alpha \leq 0.2$ . Synthetic smoke claims remain withdrawn.

## 9. Theoretical Verification

All formulas verified numerically via `experiments/verify_theory.py`.

Table 6: Theoretical guarantees verification.

| Theorem                   | Formula                                                          | Status         | Empirical Outcome                                                                                                                                                                                                        |
|---------------------------|------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 4.2 (DP radii)    | $\rho_{\text{DP},j} = \alpha / ((1-\alpha)\pi_j + \alpha)$       | ✓ Yes          | All 3 datasets $\times 5\alpha$                                                                                                                                                                                          |
| Theorem 4.3 (IF radius)   | $\rho_{\text{IF}} = 2\alpha - \alpha^2$                          | ✓ Yes          | All configurations                                                                                                                                                                                                       |
| Theorem 6.1 (guarantee)   | $(\varepsilon_{\text{DP}} + \varepsilon_{\text{IF}})$ -fairness  | <b>Partial</b> | Holds: Credit and Adult under $\tau=1$ ; LSAC/Combined wins ( $p=0.0156$ ) but LSAC/DP is degenerate (DRO loses, constant-predictor collapse at every $\alpha$ ). Pre-tau=1 Adult data ( $\tau=100$ ) showed instability |
| Remark 6.2 (monotonicity) | $\rho \rightarrow 0$ as $\alpha \rightarrow 0$ ; $\rho$ monotone | ✓ Yes          | Verified numerically                                                                                                                                                                                                     |
| Bias correction (App. F)  | $\pi_j = (\hat{\pi}_j - \alpha) / (1 - 2\alpha)$                 | ✓ Yes          | Applied in <code>_compute_radii</code>                                                                                                                                                                                   |
| Dykstra convergence       | Simplex $\cap \ell_1$ -ball                                      | ✓ Yes          | <code>max_iter=500, tol=10<sup>-5</sup></code>                                                                                                                                                                           |
| Tilted loss equivalence   | $\beta \cdot \text{logsumexp}(\ell/\beta) - \beta \log n$        | ✓ Yes          | <code>verified equivalence: True</code>                                                                                                                                                                                  |
| Algorithm 1 step order    | $\theta \rightarrow \lambda \rightarrow \tilde{p}$               | ✓ Yes          | Exact order in <code>dro_fair.py</code>                                                                                                                                                                                  |

## 10. Runtime Analysis

Table 7: Runtime comparison (seconds per experiment, CPU).

| Method     | Time (s)          | Overhead     |
|------------|-------------------|--------------|
| Naive-FAIR | $10.6 \pm 9.1$    | $1.0\times$  |
| DRO-FAIR   | $397.6 \pm 258.5$ | $37.5\times$ |

DRO-FAIR is  $\approx 37.5\times$  slower than Naive-FAIR on CPU due to  $K = 10$  inner PGD steps + Dykstra projection per epoch. The paper’s  $\approx 12\times$  is GPU-based; full-batch CPU  $k$ -NN graph construction is the dominant overhead.

## 11. Ablation Study

Table 8: Ablation study on Adult (adversarial corruption, 6 seeds).

| Variant           | $\alpha=0.2$ Acc | DP     | IF     | $\alpha=0.3$ Acc | DP     | IF     |
|-------------------|------------------|--------|--------|------------------|--------|--------|
| Standard ML       | 0.822            | 0.1034 | 0.0233 | 0.816            | 0.1135 | 0.0248 |
| Naive-FAIR        | 0.821            | 0.0905 | 0.0164 | 0.786            | 0.0339 | 0.0258 |
| DRO Joint (DP+IF) | 0.788            | 0.0484 | 0.0067 | 0.606            | 0.0109 | 0.0028 |
| DRO DP-only       | 0.803            | 0.0726 | 0.0108 | 0.647            | 0.0599 | 0.0119 |
| DRO IF-only       | 0.821            | 0.0956 | 0.0174 | 0.787            | 0.0295 | 0.0280 |

- **Standard ML:** highest accuracy ( $\approx 83\%$ ) but worst DP ( $\approx 0.175$ ) — confirms need for fairness constraints.
- **DP-only vs. Joint:** DP-only causes IF regression vs. joint at  $\alpha = 0.2$  — constraints interact.
- **IF-only:** more stable at  $\alpha = 0.3$  (avoids  $\lambda_{DP}$  runaway) — IF constraint alone does not destabilise.
- **Joint DRO at  $\alpha = 0.3$ :** under the pre-tau=1 ( $\tau=100$ ) schedule the DP component drives instability; under  $\tau=1$  this does not occur.

## 12. Week 2: Adversarial Fairness Attacks

We extend the evaluation to **gradient-based adversarial attacks** on fairness metrics, where the attacker computes the exact gradient of the fairness objective (DP or IF) with respect to each sample’s label and flips the  $\alpha n$  worst samples.

### 12.1 Fairness-Targeted PGD Attack

**DP-gradient:** For each sample  $i$  in protected group  $g$ , the gradient  $d(\lambda_{DP})/d(y_i)$  is  $+1$  if flipping increases the group rate gap and  $-1$  otherwise.

**IF-gradient:** For each sample  $i$ , we build a  $k$ -NN graph within its protected group. If  $i$ ’s label agrees with  $k_{\text{eff}}$  of its neighbors, flipping creates  $k_{\text{eff}}$  new disagreements; the gradient is  $(k_{\text{eff}} - \text{disagree})/k_{\text{eff}}$ .

**Combined:** Equal-weighted sum of DP and IF gradients.

## 12.2 Results (540 experiments, 6 seeds)

Table 9: Fairness-targeted PGD attacks: DRO vs Naive (mean over 6 seeds,  $\tau=1$  canonical).

| Dataset | Attack   | $\alpha$ | DP Naive | DP DRO | Reduction | $p$ -value |
|---------|----------|----------|----------|--------|-----------|------------|
| Adult   | DP       | 0.0      | 0.1491   | 0.1426 | 4.3%      | 0.016      |
| Adult   | DP       | 0.1      | 0.2026   | 0.1999 | 1.3%      | 0.031      |
| Adult   | DP       | 0.2      | 0.2452   | 0.2334 | 4.8%      | 0.016      |
| Adult   | DP       | 0.3      | 0.2848   | 0.2614 | 8.2%      | 0.016      |
| Adult   | DP       | 0.4      | 0.3140   | 0.2855 | 9.1%      | 0.016      |
| Adult   | IF       | 0.0      | 0.1491   | 0.1426 | 4.3%      | 0.016      |
| Adult   | IF       | 0.1      | 0.0819   | 0.0770 | 6.1%      | 0.016      |
| Adult   | IF       | 0.2      | 0.0474   | 0.0455 | 4.0%      | 0.016      |
| Adult   | IF       | 0.3      | 0.0227   | 0.0241 | -6.2%     | 0.891      |
| Adult   | IF       | 0.4      | 0.0042   | 0.0039 | 7.8%      | 0.500      |
| Adult   | COMBINED | 0.0      | 0.1491   | 0.1426 | 4.3%      | 0.016      |
| Adult   | COMBINED | 0.1      | 0.1509   | 0.1432 | 5.1%      | 0.016      |
| Adult   | COMBINED | 0.2      | 0.1963   | 0.1784 | 9.1%      | 0.016      |
| Adult   | COMBINED | 0.3      | 0.2176   | 0.1922 | 11.6%     | 0.016      |
| Adult   | COMBINED | 0.4      | 0.2110   | 0.1815 | 14.0%     | 0.016      |
| Credit  | DP       | 0.0      | 0.0127   | 0.0119 | 6.3%      | 0.016      |
| Credit  | DP       | 0.1      | 0.0151   | 0.0134 | 11.3%     | 0.016      |
| Credit  | DP       | 0.2      | 0.0198   | 0.0178 | 9.9%      | 0.016      |
| Credit  | DP       | 0.3      | 0.0253   | 0.0228 | 9.9%      | 0.016      |
| Credit  | DP       | 0.4      | 0.0191   | 0.0170 | 10.8%     | 0.016      |
| Credit  | IF       | 0.0      | 0.0127   | 0.0119 | 6.3%      | 0.016      |
| Credit  | IF       | 0.1      | 0.0056   | 0.0049 | 12.2%     | 0.109      |
| Credit  | IF       | 0.2      | 0.0053   | 0.0045 | 16.5%     | 0.031      |
| Credit  | IF       | 0.3      | 0.0049   | 0.0038 | 21.8%     | 0.016      |
| Credit  | IF       | 0.4      | 0.0049   | 0.0035 | 27.3%     | 0.047      |
| Credit  | COMBINED | 0.0      | 0.0127   | 0.0119 | 6.3%      | 0.016      |
| Credit  | COMBINED | 0.1      | 0.0117   | 0.0106 | 9.1%      | 0.031      |
| Credit  | COMBINED | 0.2      | 0.0135   | 0.0121 | 10.3%     | 0.016      |
| Credit  | COMBINED | 0.3      | 0.0177   | 0.0149 | 15.9%     | 0.016      |
| Credit  | COMBINED | 0.4      | 0.0167   | 0.0131 | 21.9%     | 0.016      |
| Lsac    | DP       | 0.0      | 0.1447   | 0.1829 | -26.4%    | 1.000      |
| Lsac    | DP       | 0.1      | 0.2201   | 0.2539 | -15.4%    | 1.000      |
| Lsac    | DP       | 0.2      | 0.1827   | 0.2230 | -22.1%    | 1.000      |
| Lsac    | DP       | 0.3      | 0.1827   | 0.2220 | -21.5%    | 1.000      |
| Lsac    | DP       | 0.4      | 0.1827   | 0.2211 | -21.0%    | 1.000      |
| Lsac    | IF       | 0.0      | 0.1447   | 0.1829 | -26.4%    | 1.000      |
| Lsac    | IF       | 0.1      | 0.0536   | 0.0614 | -14.6%    | 1.000      |
| Lsac    | IF       | 0.2      | 0.0419   | 0.0604 | -44.1%    | 1.000      |
| Lsac    | IF       | 0.3      | 0.0512   | 0.0564 | -10.2%    | 1.000      |
| Lsac    | IF       | 0.4      | 0.0502   | 0.0500 | 0.5%      | 0.500      |
| Lsac    | COMBINED | 0.0      | 0.1447   | 0.1829 | -26.4%    | 1.000      |
| Lsac    | COMBINED | 0.1      | 0.1437   | 0.1361 | 5.3%      | 0.016      |
| Lsac    | COMBINED | 0.2      | 0.1321   | 0.1248 | 5.6%      | 0.031      |
| Lsac    | COMBINED | 0.3      | 0.1874   | 0.1602 | 14.5%     | 0.016      |
| Lsac    | COMBINED | 0.4      | 0.2151   | 0.1830 | 14.9%     | 0.016      |

## 12.3 Key Findings

- **At fixed  $\tau=1$ : DRO wins on Adult and Credit DP under the DP-targeted and Combined attacks at  $\alpha \leq 0.2$  (headline).** At  $\alpha \leq 0.2$  DRO beats Naive under both attacks (Wilcoxon one-sided  $p < 0.05$ ,  $n=6$  seeds; verified in `results/canonical_tau1.json`; Adult/DP  $\alpha=0.1$  is 5/6). Accuracy is equal-or-better. At  $\alpha \geq 0.3$  both methods fall below

the constant-predictor baseline on Adult and Credit only, so no method claim is made there (LSAC/DP is pinned at majority accuracy, not below).

- **LSAC/DP is a degenerate negative result; LSAC/Combined wins:** On LSAC the model collapses to the constant (majority-class) predictor at every  $\alpha$  (accuracy  $\approx 0.902$ ), and DRO loses to Naive on LSAC/DP at 0/6 seeds. LSAC/Combined is a genuine clean win ( $p=0.0156$  at  $\alpha=0.1, 0.3, 0.4$ ). We report this openly rather than hiding it.
- **Adversarial  $\gg$  random (qualitative):** Historical pilot only; multiplicative factors not re-locked under the 540-row protocol.
- **IF attack is complete:** The IF-attack rows of the canonical grid are generated (180/180); earlier IF metrics were degenerate ( $\approx 10^{-10}$ ). Honest read: Adult/Credit  $\alpha \leq 0.2$  support DRO on DP under IF attack; LSAC/IF and Adult IF  $\alpha \geq 0.3$  do not. Canonical IF rows use fixed  $k=5$  (full  $k$  ablation not in the locked science file).
- **UTKFace (complete REAL 90/90 grid; mixed clean-test story):** Real ResNet features ( $N=23,705$ ) and full protocol (results/utkface\_canonical.json, all data\_provenance=REAL). Clean-test DP is mixed: significant DRO wins mainly at high  $\alpha$  (e.g. DP/Combined  $\alpha=0.4$ , 6/6,  $p=0.016$ ), not a copy of Adult/Credit  $\alpha \leq 0.2$ . Synthetic smoke “DRO inverts on image features” remains withdrawn.

### 13. Conclusion

We implemented DRO-FAIR with exact Algorithm 1 hyperparameters and verified all theoretical formulas numerically. Our adversarial corruption extension — PGD feature attacks, coordinated label flips, minority-targeted attribute flips — provides a much harder evaluation than the paper’s random noise at the same  $\alpha$ .

The headline finding: **fixing the prediction temperature at  $\tau=1$  across all  $\alpha$  makes DRO-FAIR achieve significantly lower DP than Naive-FAIR on Adult and Credit at  $\alpha \leq 0.2$  under the DP-targeted and Combined attacks** (verified in results/canonical\_tau1.json; e.g. Adult  $\alpha=0.2$ : Naive 0.245 vs DRO 0.233), with accuracy equal-or-slightly-better. The earlier “DRO is fragile / DRO loses on Adult” result was a high-temperature artifact (stepped  $\tau=100$  schedule for  $\alpha \leq 0.3$ ); switching to fixed  $\tau=1$  inverts that conclusion. LSAC/DP is a degenerate negative result (constant-predictor collapse, DRO loses at 0/6 seeds); LSAC/Combined is a genuine win ( $p=0.0156$ ). The IF-attack rows are complete; under IF attack Adult/Credit  $\alpha \leq 0.2$  still support DRO on DP while LSAC/IF is a DP loss (mixed; earlier pre-fix IF metrics were degenerate,  $\approx 10^{-10}$ ).

On Adult, a historical pilot found adversarial PGD raises DP far more than matched- $\alpha$  random label noise (qualitative; not a locked multiplicative factor), which still rules out a pure “attack too weak” counter-explanation for the low- $\alpha$  PGD cells.

The canonical full grid is complete (540 rows); Adult and Credit at  $\alpha \leq 0.2$  are the mature findings. LSAC/Combined is a genuine win ( $p=0.0156$ ); LSAC/DP is a degenerate negative result. The IF-attack rows are complete (180/180; Adult/Credit  $\alpha \leq 0.2$  support DRO on DP; LSAC/IF is a DP loss — not a three-attack sweep). UTKFace REAL grid is complete (90/90; clean-test DP mixed — significant DRO wins mainly at high  $\alpha$ ; not an Adult copy). Prior synthetic smoke tests withdrawn. Q5 empirical radii math in appendix (see also docs/KULDEEP\_CORRECTION.md).

**Practical takeaway.** When deploying DRO-FAIR, fix the prediction temperature and re-validate rather than relying on a  $\tau$ -schedule. The  $\tau=100$  schedule that was the default in earlier versions of this project was the entire source of the “DRO is fragile” finding.

**Future directions.**

- IF-attack third of the canonical grid — DONE (180/180 after metric fix; honest mixed read).
- UTKFace REAL feature pilot — DONE (90/90; mixed clean-test). Optional later: flair2 pixel-space if greenlit.
- Empirical radii calibration under known attack structure (Q5; uniform-vs-empirical comparison pending canonical\_tau1\_empirical.json).
- Dataset-adaptive  $\lambda_{\max}$ .

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Code: [github.com/Srujan0798/DRO-FairML](https://github.com/Srujan0798/DRO-FairML) | Canonical  $\tau=1$  grid complete (540 rows; results/canonical\_tau1.json)  
 UTKFace REAL 90/90, 62 unit tests, theory verifications.

**A. Q5 Derivation: Empirical  $\pi_{\text{clean}}$  under Known Coordinated Attack**


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**Source.** Derivation extracted from the implementation docstring in `src/training/dro_fair.py:_empirical_pi_clean` (B's theory writeup per MASTER\_PLAN §5). Used only when `radii_mode='empirical'` and the attack structure is known a-priori (70% of corruption budget targets the minority group). DRO receives only  $\alpha$  and this known structure; never the true per-sample corruption mask (global constraint §0).

**Setup.** Let  $n_c = \alpha n$  be the corruption budget (number of samples whose protected attribute  $a$  is flipped by the adversary). Under the coordinated attack:

- 70% of the budget ( $0.7n_c$ ) is applied to samples whose *clean* label is the minority group: their  $a$  is flipped from minority  $\rightarrow$  majority.
- 30% of the budget ( $0.3n_c$ ) is applied to samples whose *clean* label is the majority group: their  $a$  is flipped from majority  $\rightarrow$  minority.

**Observed counts after attack.** Let  $N_{\min}$  and  $N_{\text{maj}}$  be the *clean* counts. Post-attack observed counts:

$$\begin{aligned} N_{\min, \text{obs}} &= N_{\min} - 0.7n_c + 0.3n_c = N_{\min} - 0.4n_c \\ N_{\text{maj}, \text{obs}} &= N_{\text{maj}} - 0.3n_c + 0.7n_c = N_{\text{maj}} + 0.4n_c \end{aligned}$$

Dividing by  $n$ :

$$\begin{aligned} \pi_{\text{obs}}[\text{min}] &= \pi_{\text{clean}}[\text{min}] - 0.4\alpha \\ \pi_{\text{obs}}[\text{maj}] &= \pi_{\text{clean}}[\text{maj}] + 0.4\alpha \end{aligned}$$

**Inversion (recover clean proportions):**

$$\begin{aligned} \pi_{\text{clean}}[\text{min}] &= \pi_{\text{obs}}[\text{min}] + 0.4\alpha \\ \pi_{\text{clean}}[\text{maj}] &= \pi_{\text{obs}}[\text{maj}] - 0.4\alpha \end{aligned}$$

Clip to  $[0, 1]$  and renormalize (the code path does exactly this; see `_empirical_pi_clean`). The minority index is taken as  $\arg \min(\pi_{\text{obs}})$  (post-attack observed minority; for  $\alpha \leq 0.4$  this matches the clean minority on our datasets).

**Validity conditions.** Exact for pure attribute-flip coordinated 70/30 attack provided  $\alpha n < N_{\min}$  (no clipping of the 70% allocation). Holds for  $\alpha \leq 0.4$  on Adult/Credit/LSAC. When

`a_val` (clean validation attributes) is available it is preferred; empirical mode is the fallback exploiting known attack structure only.

**Why this belongs in the paper.** Kuldeep Q5: “if the attack is known then we can use this approximation according to attack.” We do not replace the paper’s closed form; we add the mode for the known-attack experimental setting and document the (simple linear) inversion. Uniform mode remains default for blind/unknown-attack scenarios.

## References

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